

A maximal semantic lattice identity: Conjecture 2610 is false

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The source defines strength semantically, by classes of lattices satisfying identities, and asserts that every lattice identity has a strictly stronger one. Consider the identity

$$e : \quad x_0 = x_1.$$

Variables themselves are lattice terms, so this is a lattice identity.

Theorem 1. *No lattice identity is strictly stronger than e in the asserted semantic sense.*

Proof. If a nonempty lattice satisfies e under every assignment, assign arbitrary elements a, b to x_0, x_1 . This gives $a = b$. Thus the models of e are exactly the one-element lattices. Conversely a one-element lattice satisfies every lattice identity, since any two evaluated terms have the same value. Consequently, for every proposed new identity f , every model of e is already a model of f . Its model class cannot be a proper subclass of the model class of e . This rules out a strict refinement. $\square$

The identity is consistent: the one-element lattice is an explicit model. Both language versions quantify over every identity and impose no requirement of a nontrivial model. Restricting identities to those admitting a nontrivial lattice would be a different assertion. This counterexample already refutes the necessary “every identity” clause, so the later proposed doubling construction cannot repair the stated conjunction.

Formal verification. `LatticeModel` has a nonempty carrier, meet and join, and the full commutativity, associativity, idempotence, and absorption axioms. These are the usual equational lattice axioms. `Term` is the inductive type of all finite variable/meet/join terms; `Satisfies` quantifies over all assignments. `StrictlyStronger new old` requires inclusion of the classes of models and a lattice satisfying the old identity but not the new one. Thus the formal strictness is proper semantic inclusion, not increased formula length. The universe-polymorphic theorem `collapse_implies_every_identity` treats arbitrary lattices and arbitrary new terms, not just a finite list. It implies `collapse_has_no_strict_refinement` and the final `conjecture2610_false`. The concrete one-element model and its satisfaction proof are also checked.

Lean 4.19.0 with `Std` checks the file. The final negation depends only on the standard propositional extensionality axiom; no unproved axioms or evaluation oracles are introduced.